

* Wi-fi

- Wi-fi is a wireless networking technology that allows devices to connect to a local area network using radio waves based on IEEE 802.11 standards.

* Access point (AP)

- An access point is a networking device that connects wireless devices to a wired or wireless network and manages wireless communication.

* Wireless client:

- Any device such as a PC, laptop or smartphone that connects to a network using wi-fi.

* Packet:

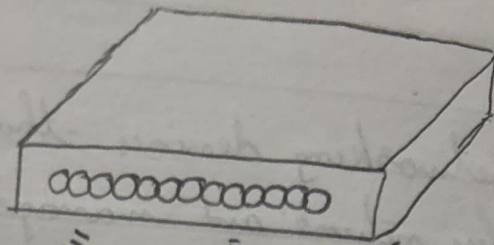
- A packet is a small unit of data transmitted over a network.

* ICMP

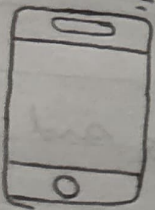
- ICMP is a network protocol used for error reporting and testing, commonly used by the ping command.

* Packet capture

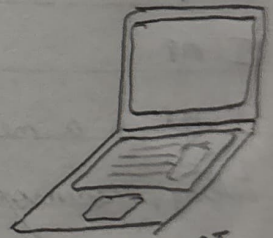
- It is the process of monitoring and recording packets transmitted over a network for analysis.



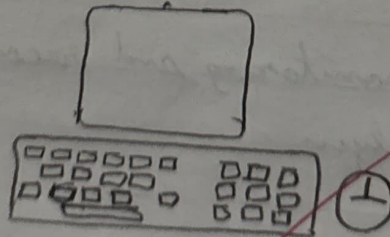
Access point - PT



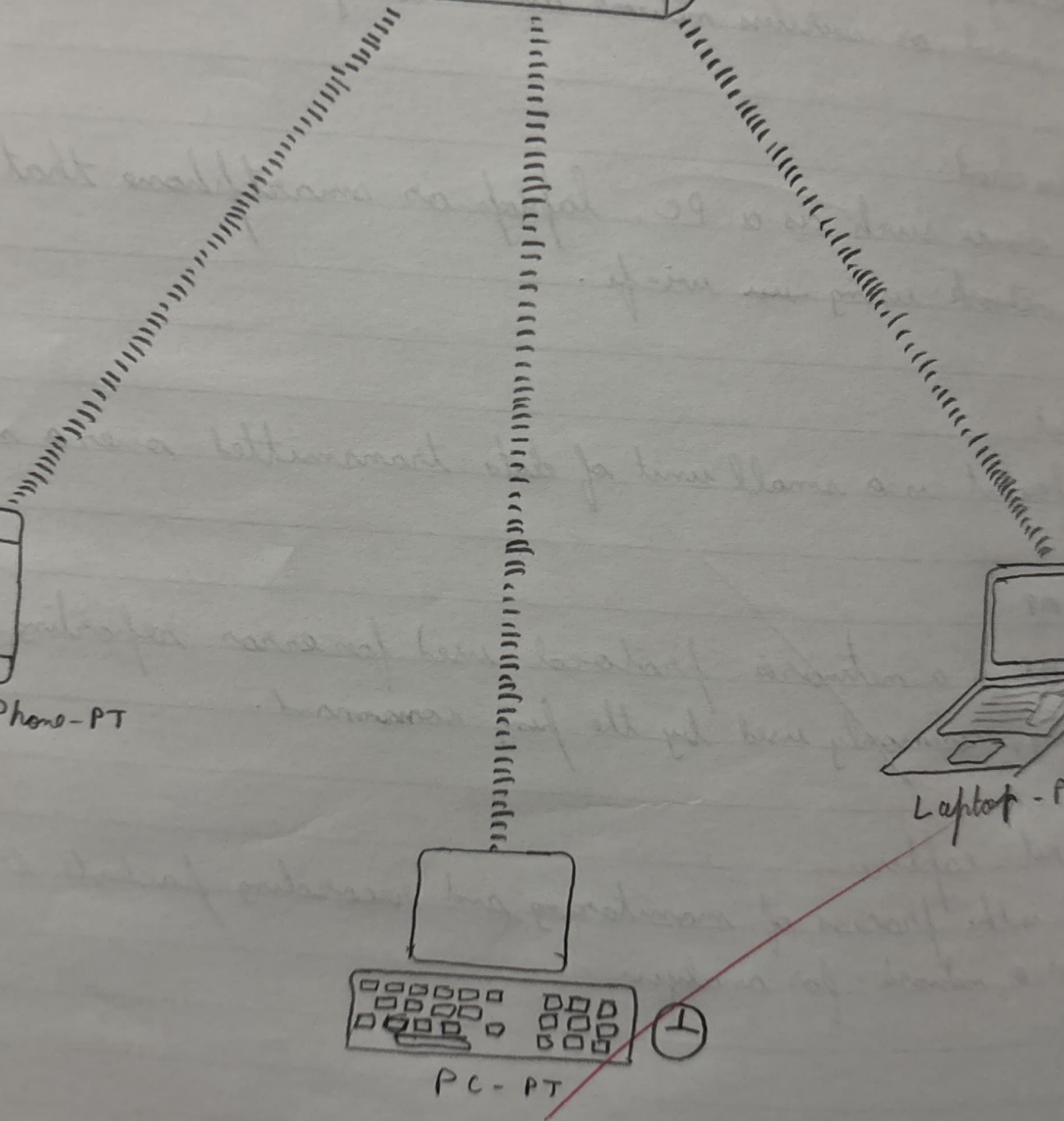
Smart Phone - PT



Laptop - PT



PC - PT



* Advantages.

- provides wireless mobility
- Reduces labelling cost
- Supports multiple devices
- Flexible and scalable network.

* Disadvantages

- Limited range compared to wired networks
- Susceptible to interference
- Lower speed than wired Ethernet.
- Performance depends on signal strength.

* Applications.

- Home and office networking
- Internet access for mobile devices
- Educational institutions
- Public hotspots.

* Devices used

- Access point - PT
- PC - PT
- Laptop - PT
- Smartphone - PT
- Cisco packet Tracer

* Procedure

- In cisco packet tracer drag & place one AP on workspace.
- Add wireless devices, one PC-PT, one laptop-PT, smartphone-PT.
- Ensure all devices are ~~inserted~~ within the wireless range of the access point.
- Configure the access point with default settings.
- Switch to ~~the~~ simulation mode.
- Observe packet movement & status in the event.
- Verify successful packet transmission.

* Observation

- ICMP packets were transmitted successfully.
- Packet status showed "Successful".
- Wireless communication was observed without using physical cables.

* Results.

- The study and demonstration of wi-fi communication were successfully carried out using cisco packet tracer.
- Packet capture confirmed successful ICMP communication between wireless devices.

* Conclusion

- This experiment works of a wi-fi network, where multiple wireless devices communicate through an AP.
- Packet capture shows successful data transmission proving that wi-fi provides reliable & flexible wireless communication for LANs.